

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials.
- Show all of your work, and use correct notation. Failure to do these things will result in a deduction of points.
- You have **1 hour** to complete this exam.
- You've got this—I believe in you!

1. (22 pts.) Suppose $Y_1, Y_2, \dots, Y_6 \stackrel{i.i.d.}{\sim} N(0, 1)$.

(a.) (4 pts.) What is the distribution of $\bar{Y} = \frac{1}{6} \sum_{i=1}^6 Y_i$?

Solution: Chapter 7, Theorem 7.1.

$$N(0, 1/\sqrt{6})$$

(b.) (4 pts.) What is the distribution of $\sum_{i=1}^6 Y_i^2$?

Solution: Chapter 7, Theorem 7.2.

$$\chi_6^2$$

(c.) (10 pts.) Find an interval that will contain $S^2 = \frac{1}{5} \sum_{i=1}^6 (Y_i - \bar{Y})^2$ with probability .99.

Solution: Similar to Chapter 7 notes, “Example 2 cont’d.”

WTF: a and b such that $P(a \leq S^2 \leq b) = .99$.

$$\begin{aligned} P(a \leq S^2 \leq b) = .99 &\implies P\left(5a \leq \frac{(n-1)S^2}{\sigma^2} \leq 5b\right) = .99 \\ &\implies P\left(5a \leq W \leq 5b\right) = .99, \end{aligned}$$

where $W = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \equiv \chi_5^2$.

Thus, $P(W < 5a) = .005$, which implies that $5a = \chi_{.995, df=5}^2 = .411740$ (Table 6); i.e., $a = .411740/5 = \mathbf{.0823}$.

Similarly, $P(W > 5b) = .005$, which implies that $5b = \chi_{.005, df=5}^2 = 16.7496$ (Table 6); i.e., $b = 16.7496/5 = \mathbf{3.3499}$.

Hence, an interval that contains S^2 with probability .99 is $(.0823, 3.3499)$.

(d.) (4 pts.) What is the distribution of $\bar{Y}/(S/\sqrt{6})$?

Solution: Chapter 7, Fact 5.

t_5

2. (26 pts.) Consider $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Uniform}(\theta, \theta + 6)$. Suppose we wish to estimate θ . Consider the estimator $\hat{\theta} = \bar{Y} + 3$.

(a.) (6 pts.) Calculate $E(\hat{\theta})$.

Solution: Similar to Section 8.2 notes, Example 1.

$$E(\hat{\theta}) = E(\bar{Y} + 3) = E(\bar{Y}) + 3 = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) + 3 = \frac{\theta + (\theta + 6)}{2} + 3 = \theta + 6$$

(b.) (6 pts.) Calculate $B(\hat{\theta})$.

Solution: Similar to Section 8.2 notes, Example 1.

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta = (\theta + 6) - \theta = 6$$

(c.) (8 pts.) Calculate $MSE(\hat{\theta})$.

Solution: Similar to Section 8.2 notes, Example 1.

$$MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2 = \frac{3}{n} + 36$$

$$\bullet V(\hat{\theta}) = V(\bar{Y} + 3) = V(\bar{Y}) = V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n^2} \sum_{i=1}^n V(Y_i) = \frac{((\theta+6)-\theta)^2}{12n} = \frac{3}{n}$$

(d.) (6 pts.) Modify $\hat{\theta}$ to get an unbiased estimator of θ . Call this new estimator $\hat{\theta}^*$.

Solution: Similar to Section 8.2 notes, Example 1.

Note that $E(\hat{\theta}) = \theta + 6$, so let's try $\hat{\theta}^* = \hat{\theta} - 6 = (\bar{Y} + 3) - 6 = \bar{Y} - 3$.

Then, $E(\bar{Y} - 3) = E(\bar{Y}) - 3 = (\theta + 3) - 3 = \theta$.

3. (22 pts.) Suppose that Y is a single observation from an exponential distribution with mean θ .

- (a.) (10 pts.) Using the method of moment-generating functions, find the distribution of $U = \frac{Y}{2\theta}$.

Solution: Similar to Section 8.5 notes, Example 4.

$$m_U(t) = E(e^{tU}) = E(e^{(t/(2\theta))Y}) = m_Y(t/(2\theta)) = \left(1 - \theta\left(\frac{t}{2\theta}\right)\right)^{-1} = \left(1 - \frac{t}{2}\right)^{-1},$$

which is the mgf of an $Exp(1/2)$ distribution. Hence, by uniqueness of mgfs, $U \sim Exp(1/2)$.

- (b.) (12 pts.) Using U as your pivotal quantity, find a 95% confidence interval for θ .

Solution: Similar to Section 8.5 notes, Example 4.

WTF: a and b such that $P(a \leq U \leq b) = .95$. Thus, we can choose a and b such that $P(U < a) = .025$ and $P(U > b) = .025$.

$$\begin{aligned} P(U < a) = .025 &\implies \int_0^a 2e^{-2u} du = .025 \\ &\implies -e^{-2u} \Big|_0^a = .025 \\ &\implies -e^{-2a} + 1 = .025 \\ &\implies a \approx .0127. \end{aligned}$$

$$\begin{aligned} P(U > b) = .025 &\implies \int_b^\infty 2e^{-2u} du = .025 \\ &\implies -e^{-2u} \Big|_b^\infty = .025 \\ &\implies 0 + e^{-2b} = .025 \\ &\implies b \approx 1.8444. \end{aligned}$$

So, we have that $P(.0127 \leq U \leq 1.8444) = .95$. We next need to isolate θ in the middle.

$$\begin{aligned}
P\left(.0127 \leq \frac{Y}{2\theta} \leq 1.8444\right) &= .95 \implies P\left(\frac{1}{.0127} \geq \frac{2\theta}{Y} \geq \frac{1}{1.8444}\right) = .95 \\
&\implies P\left(\frac{Y}{3.6888} \leq \theta \leq \frac{Y}{.0254}\right) = .95.
\end{aligned}$$

Hence, a 95% CI is $\left(\frac{Y}{3.6888}, \frac{Y}{.0254}\right)$.

4. (18 pts.) A wildlife researcher is studying the population of a certain species of frogs in a large nature reserve. She randomly selects a sample of 200 frogs and finds that 34 of them have a specific skin disease.
- (a.) (10 pts.) Calculate a 90% confidence interval for the true proportion of frogs in this reserve that have this skin disease. You may assume the conditions needed to construct a large-sample confidence interval are satisfied (they are ☺).

Solution: Similar to Section 8.6 notes, Example 9.

$$\begin{aligned}\hat{p} \pm z_{.05} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} &= \frac{34}{200} \pm 1.645 \sqrt{\frac{(\frac{34}{200})(1 - \frac{34}{200})}{200}} \\ &= .17 \pm .0437 \\ &= (.1263, .2137)\end{aligned}$$

- (b.) (8 pts.) Interpret your interval in context of the problem.

Solution: Similar to Section 8.6 notes, Example 9.

We are 90% confident that the true proportion of frogs in this reserve that have this skin disease is between .1263 and .2137.

Choose **one** of problems 5-7 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may randomly select which one to grade.

5. (12 pts.) Prove that $MSE(\hat{\theta})$, which is equal to $E[(\hat{\theta} - \theta)^2]$, can be expressed as

$$MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2.$$

Solution: Proof of Fact from Section 8.2 notes.

$$\begin{aligned} MSE(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] \\ &= E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2] \\ &= E[(\hat{\theta} - E(\hat{\theta}))^2 + 2B(\hat{\theta})(\hat{\theta} - E(\hat{\theta})) + (B(\hat{\theta}))^2] \\ &= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2B(\hat{\theta})(E(\hat{\theta}) - E(\hat{\theta})) + E[(B(\hat{\theta}))^2] \\ &= V(\hat{\theta}) + (B(\hat{\theta}))^2 \end{aligned}$$

6. (12 pts.) Suppose that $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $V(\hat{\theta}_1) = \sigma_1^2 > 0$, and $V(\hat{\theta}_2) = \sigma_2^2 > 0$. Consider the unbiased estimator $\hat{\theta}_3 = a\hat{\theta}_1 + (1-a)\hat{\theta}_2$, where a is a constant. If $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, what value of a should be chosen to minimize the variance of $\hat{\theta}_3$? Verify that this value is, indeed, a minimizer.

Solution: HW problem 8.6b.

$$V(\hat{\theta}_3) = V[a\hat{\theta}_1 + (1-a)\hat{\theta}_2] = a^2V(\hat{\theta}_1) + (1-a)^2V(\hat{\theta}_2) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2,$$

where the second equality is due to independence.

$$\frac{dV(\hat{\theta}_3)}{da} = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 \stackrel{set}{=} 0 \implies a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

To verify that this value of a is a minimizer, we should check that the second derivative is positive.

$$\frac{d^2V(\hat{\theta}_3)}{da^2} = 2\sigma_1^2 + 2\sigma_2^2 > 0.$$

7. (12 pts.) Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Prove that $T \sim t_{n-1}$. You may reference the following fact in your proof:

Fact: Suppose $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$. Then, if Z and W are independent,

$$T = \frac{Z}{\sqrt{W/\nu}} \sim t_\nu.$$

Solution: Proof of Fact 5 from Section 7.2 notes.

Suppose $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Define $Z = \sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right) \sim N(0, 1)$ and $W = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$. Note that Z and W are independent.

Then, using the provided Fact,

$$T = \frac{Z}{\sqrt{\frac{W}{n-1}}} = \frac{\sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right)}{\sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right) \sim t_{n-1}.$$